

Z-Test Explanation

The **Z-test** is a type of hypothesis test used to determine if the means of two data sets are significantly different from each other, or if a single sample mean is significantly different from a hypothesized population mean. It is specifically based on the **Normal Distribution**.



When to Apply the Z-Test

You should use a Z-test when you can satisfy one of the following two primary conditions:

Condition	Description	Rationale
1. Large Sample Size	The sample size (n) is greater than or equal to 30 ($n \geq 30$).	The Central Limit Theorem states that for large samples, the distribution of the sample mean approaches the Normal Distribution, regardless of the population's original shape.
2. Known Population Standard Deviation	The true standard deviation of the population (σ) is known , even if the sample size is small.	When σ is known, you use the standard normal distribution (Z-distribution) for the test. (In reality, this is rare, making the large sample size rule the most common reason for using the Z-test).

// **Contrast with T-test:** If the sample size is small ($n < 30$) AND the population standard deviation (σ) is unknown, you must use the **T-test**, which relies on the t -distribution to account for the added uncertainty.



How to Apply the Z-Test (One-Sample Mean)

The application of the Z-test follows the general procedure for hypothesis testing. Here is the process for testing a single population mean.

1. State the Hypotheses

- **Null Hypothesis (H_0):** The population mean (μ) is equal to the hypothesized value (μ_0). $H_0 : \mu = \mu_0$.
- **Alternate Hypothesis (H_1):** The population mean is different from the hypothesized value. This determines the type of test:
 - **Two-tailed:** $H_1 : \mu \neq \mu_0$
 - **Right-tailed:** $H_1 : \mu > \mu_0$
 - **Left-tailed:** $H_1 : \mu < \mu_0$

2. Choose the Level of Significance (α)

This is the maximum risk of rejecting a true null hypothesis, commonly set at $\alpha = 0.05$ (5%) or $\alpha = 0.01$ (1%).

3. Calculate the Test Statistic ($Z_{\text{Calculated}}$)

The Z-test statistic measures the difference between your sample mean ($\bar{x}$) and the hypothesized population mean (μ) in units of the standard error.

The formula for the **One-Sample Z-test** is:

$$Z_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$: Sample Mean
- μ_0 : Hypothesized Population Mean (from H_0)

- σ : Population Standard Deviation (Use the Sample Standard Deviation S if σ is unknown and $n \geq 30$)
- n : Sample Size

4. Determine the Critical Value (Z_{Critical})

Look up the critical Z -value in the Standard Normal Distribution table based on your chosen α and the type of test (one or two-tailed).

α	Two-Tailed ($Z = \pm$)	One-Tailed (Z only)
0.05 (5%)	± 1.96	± 1.645
0.01 (1%)	± 2.58	± 2.33

5. Make a Conclusion (Critical Value Approach)

Compare the absolute value of $Z_{\text{Calculated}}$ to the absolute value of Z_{Critical} .

- **Reject H_0 :** If $|Z_{\text{Calculated}}| > |Z_{\text{Critical}}|$, the result is **statistically significant**, and the difference is too large to be explained by chance.
- **Accept H_0 (Fail to Reject H_0):** If $|Z_{\text{Calculated}}| \leq |Z_{\text{Critical}}|$, the result falls into the acceptance region, and there is not enough evidence to reject the claim ($\mu = \mu_0$).

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What is P-Value

The **P-value** (Probability Value) is an alternative way to make the final conclusion in a hypothesis test, providing a more precise measure of the evidence against the Null Hypothesis (H_0).

Simple Definition

The P-value is the probability of obtaining a test statistic **as extreme as**, or **more extreme than**, the one you calculated from your sample data, **assuming that the Null Hypothesis (H_0) is actually true**.

Interpretation

Think of the P-value as the measure of surprise:

- **A Small P-value:** Means the observed sample result is **highly unlikely** to occur if H_0 were true. This is strong evidence *against* H_0 .
- **A Large P-value:** Means the observed sample result is **likely** to occur if H_0 were true. This is weak evidence *against* H_0 .

Decision Rule (P-Value Approach)

Instead of comparing the calculated statistic to a critical value, you compare the P-value to the pre-chosen Level of Significance (α).

P-value vs. α	Decision	Explanation
P-value $\leq \alpha$	Reject the Null Hypothesis (H_0)	The chance of observing your data (if H_0 were true) is less than the risk of error (α) you were willing to accept.
P-value $> \alpha$	Accept the Null Hypothesis (H_0) (or Fail to Reject)	The chance of observing your data (if H_0 were true) is higher than the risk of error (α) you were willing to accept.

// **Example:** If you set $\alpha = 0.05$ and your test yields a P-value of 0.01: Since $0.01 \leq 0.05$, you **Reject H_0** .

I will solve the hypothesis testing examples using the steps of the **Z-test** and then explain the conclusion using both the **Critical Value Approach** and the **P-value Approach** where applicable.

Hypothesis Testing (Z-test/Large Sample Tests) Examples

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Example 1: Testing a Single Population Mean (Known σ)

Question: A random sample of **900 members** has a mean **3.4 cm**. Can it be reasonably regarded as a sample from a large population of mean **3.2 cm** and S.D. **2.3 cm**? (Test at **5% level of significance**)

1. State the Data and Hypotheses

- Sample Size: $n = 900$ (Large Sample, $n > 30$)
- Sample Mean: $\bar{x} = 3.4$ cm
- Hypothesized Population Mean: $\mu_0 = 3.2$ cm
- Population Standard Deviation: $\sigma = 2.3$ cm
- Level of Significance: $\alpha = 0.05$

Since the question asks if the sample **can be regarded as** from the population, we test for any difference (it could be greater or smaller), leading to a **Two-Tailed Test**.

- **Null Hypothesis (H_0):** $\mu = 3.2$ (The sample comes from the population; no difference.)
- **Alternate Hypothesis (H_1):** $\mu \neq 3.2$ (The sample does **not** come from the population; there is a difference.)

2. Determine the Critical Value (Two-Tailed)

For $\alpha = 0.05$ in a two-tailed Z-test, the critical values are:

$$Z_{\text{Critical}} = \pm 1.96$$

3. Calculate the Test Statistic ($Z_{\text{Calculated}}$)

The formula is $Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$:

$$Z_{\text{Calculated}} = \frac{3.4 - 3.2}{\frac{2.3}{\sqrt{900}}} = \frac{0.2}{\frac{2.3}{30}} = \frac{0.2}{0.0767} \approx 2.6076$$

4. Conclusion

- **Critical Value Approach:**
 - $|Z_{\text{Calculated}}| = 2.6076$
 - $|Z_{\text{Critical}}| = 1.96$
 - Since $2.6076 > 1.96$, the calculated Z falls in the rejection region.
- **Result: Reject H_0 .**
- **Interpretation:** The sample mean of 3.4 cm is **significantly different** from the hypothesized population mean of 3.2 cm at the 5% level. It **cannot** be reasonably regarded as a sample from that population.



Example 2: Testing a Single Population Mean (Unknown σ , Calculated from Data)

Question: An insurance agent claims the avg. age of policy holders is **less than** the avg. age of all agents, which is **30.5 yrs**. A sample of **100 policy holders** gave the following age distribution. Calculate the mean and S.D. and test his claim at **5% level of significance**.

- $n = 100$ (Large Sample, $n > 30$)
- Hypothesized Population Mean: $\mu_0 = 30.5$ years

- Claim: Average age is **less than** 30.5 years (This is a **Left-Tailed Test**).
- Level of Significance: $\alpha = 0.05$

1. Calculate Sample Mean ($\bar{x}$) and Sample Standard Deviation (S)

We need to calculate $\bar{x}$ and S from the grouped data.

Age Group	Midpoint (x)	Frequency (f)	$f \cdot x$	x^2	$f \cdot x^2$
16-20	18	12	216	324	3888
21-25	23	22	506	529	11638
26-30	28	20	560	784	15680
31-35	33	30	990	1089	32670
36-40	38	16	608	1444	23104
Total		$\sum f = 100$	$\sum fx = 2880$		$\sum fx^2 = 86980$

- **Sample Mean ($\bar{x}$):**

$$\bar{x} = \frac{\sum fx}{n} = \frac{2880}{100} = \mathbf{28.8 \text{ years}}$$

- **Sample Variance (S^2):**

$$S^2 = \frac{\sum fx^2}{n} - (\bar{x})^2 = \frac{86980}{100} - (28.8)^2 = 869.8 - 829.44 = 40.36$$

- **Sample Standard Deviation (S):**

$$S = \sqrt{40.36} \approx \mathbf{6.353 \text{ years}}$$

2. State Hypotheses and Critical Value

- **Null Hypothesis (H_0):** $\mu = 30.5$
- **Alternate Hypothesis (H_1):** $\mu < 30.5$ (Left-Tailed Test, supporting the agent's claim)

For $\alpha = 0.05$ in a left-tailed Z-test, the critical value is:

$$Z_{\text{Critical}} = -1.645$$

3. Calculate the Test Statistic ($Z_{\text{Calculated}}$)

Since n is large, we use S as an estimate for σ : $\sigma \approx S = 6.353$.

$$Z_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{28.8 - 30.5}{\frac{6.353}{\sqrt{100}}} = \frac{-1.7}{\frac{6.353}{10}} = \frac{-1.7}{0.6353} \approx -2.676$$

4. Conclusion

- **Critical Value Approach:**
 - $Z_{\text{Calculated}} = -2.676$
 - $Z_{\text{Critical}} = -1.645$
 - Since -2.676 is to the left of -1.645 , it falls in the rejection region.
- **Result: Reject H_0 .**
- **Interpretation:** The average age of the policy holders who insured through the agent ($\bar{x} = 28.8$) is **significantly less** than 30.5 years. The agent's claim is **supported** at the 5% level of significance.

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Example 3: Testing the Difference Between Two Population Means

Question: Sample 1: $n_1 = 100$ people, $\bar{x}_1 = 210$, $S_1 = 10$. Sample 2: $n_2 = 150$ persons, $\bar{x}_2 = 220$, $S_2 = 12$. Test whether there is any significance difference between the avg. income of the localities. (Test at **5% level of significance**)

1. State the Data and Hypotheses

- Sample 1 (Locality 1): $n_1 = 100$, $\bar{x}_1 = 210$, $S_1 = 10$

- Sample 2 (Locality 2): $n_2 = 150$, $\bar{x}_2 = 220$, $S_2 = 12$
- Both samples are large ($n_1, n_2 > 30$).
- Level of Significance: $\alpha = 0.05$

Since the question asks if there is **any significance difference**, we use a **Two-Tailed Test**.

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2$ or $\mu_1 - \mu_2 = 0$ (No difference in average income.)
- **Alternate Hypothesis (H_1):** $\mu_1 \neq \mu_2$ or $\mu_1 - \mu_2 \neq 0$ (There is a difference.)

2. Determine the Critical Value (Two-Tailed)

For $\alpha = 0.05$ in a two-tailed Z-test, the critical values are:

$$Z_{\text{Critical}} = \pm 1.96$$

3. Calculate the Test Statistic ($Z_{\text{Calculated}}$)

The formula for the Z-test for the difference between two means (using S_1 and S_2 as estimates for σ_1 and σ_2):

$$Z_{\text{Calculated}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Since H_0 assumes $\mu_1 - \mu_2 = 0$:

$$\begin{aligned} Z_{\text{Calculated}} &= \frac{(210 - 220) - 0}{\sqrt{\frac{10^2}{100} + \frac{12^2}{150}}} = \frac{-10}{\sqrt{\frac{100}{100} + \frac{144}{150}}} \\ Z_{\text{Calculated}} &= \frac{-10}{\sqrt{1.00 + 0.96}} = \frac{-10}{\sqrt{1.96}} = \frac{-10}{1.4} \approx -7.14 \end{aligned}$$

4. Conclusion

- **Critical Value Approach:**
 - $|Z_{\text{Calculated}}| = 7.14$
 - $|Z_{\text{Critical}}| = 1.96$

- Since $7.14 > 1.96$, the calculated Z falls in the rejection region.
- **Result: Reject H_0 .**
- **Interpretation:** There is a **highly significant difference** between the average incomes of the two localities at the 5% level.

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Hypothesis Testing (P-value Approach) Example

Question: A manufacturing process produces TV tubes with an average life $\mu = 1200$ hr and $\sigma = 300$ hrs. A new process is thought to give tubes a **higher average life**. A sample of **100 tubes** has an average life $\bar{x} = 1265$ hr. Is the new process really any better than the old?

1. State the Data and Hypotheses

- Sample Size: $n = 100$
- Sample Mean: $\bar{x} = 1265$ hr
- Hypothesized Population Mean: $\mu_0 = 1200$ hr
- Population Standard Deviation: $\sigma = 300$ hr (Known σ , so Z-test is appropriate)
- Claim: Higher average life (This is a **Right-Tailed Test**).
- **Null Hypothesis (H_0):** $\mu = 1200$ (The new process is not better.)
- **Alternate Hypothesis (H_1):** $\mu > 1200$ (The new process is better.)

2. Calculate the Test Statistic ($Z_{\text{Calculated}}$)

$$Z_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{1265 - 1200}{\frac{300}{\sqrt{100}}} = \frac{65}{\frac{300}{10}} = \frac{65}{30} \approx 2.167$$

3. Determine the P-Value

The P-value is the probability of observing a Z -score of 2.167 or greater in a standard normal distribution. This is $P(Z \geq 2.167)$.

Using the Z-table:

- The area to the left of $Z = 2.167$ is approximately 0.9845.
- The area to the right (the P-value) is:

$$P\text{-value} = 1 - 0.9845 = \mathbf{0.0155}$$

4. Conclusion (P-Value Approach)

The problem implicitly uses $\alpha = 0.05$ for comparison.

- **Compare P-value to α :**
 - $P\text{-value} = 0.0155$
 - $\alpha = 0.05$
 - Since $0.0155 \leq 0.05$, the **P-value is less than α** .
- **Result: Reject H_0 .**
- **Interpretation:** The probability of getting a sample mean of 1265 (or higher) by chance, if the true mean were still 1200, is very low (1.55%). Because 1.55% is less than the accepted risk of 5%, we conclude that the new process **is really better** than the old one.

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Quiz and Multiple Choice Questions Solutions

Quiz Question 1: Could the mean weight in the population be 120 pounds?

- $n = 800$, $\bar{x} = 112$, $S = 20$, $\mu_0 = 120$, $\alpha = 0.05$ (Two-Tailed $Z_{\text{Crit}} = \pm 1.96$)

1. **Hypotheses:** $H_0 : \mu = 120$ vs. $H_1 : \mu \neq 120$.

2. $Z_{\text{Calculated}}$:

$$Z = \frac{112 - 120}{\frac{20}{\sqrt{800}}} = \frac{-8}{\frac{20}{28.28}} = \frac{-8}{0.707} \approx -11.315$$

3. **Conclusion:** $|-11.315| > 1.96$. **Reject** H_0 .

4. **Answer:** No, the sample mean of 112 pounds is significantly different from 120 pounds.

Quiz Question 2: Could the mean weight in the population be 120 pounds?

• $n = 1000$, $\bar{x} = 112$, $S = 20$, $\mu_0 = 120$, $\alpha = 0.05$ (Two-Tailed $Z_{\text{Crit}} = \pm 1.96$)

1. **Hypotheses:** $H_0 : \mu = 120$ vs. $H_1 : \mu \neq 120$.

2. $Z_{\text{Calculated}}$:

$$Z = \frac{112 - 120}{\frac{20}{\sqrt{1000}}} = \frac{-8}{\frac{20}{31.62}} = \frac{-8}{0.632} \approx -12.65$$

3. **Conclusion:** $|-12.65| > 1.96$. **Reject** H_0 .

4. **Answer:** No, the sample mean of 112 pounds is significantly different from 120 pounds. (Note: Increasing the sample size from 800 to 1000 made the difference even more statistically significant.)

Quiz Question 3: Random sample of 900 members has mean 3.4cms. Population mean 3.2cms and S.D. 2.3 cms. Tabulated value 1.96 at 5%.

This question is identical to **Example 1**.

• $Z_{\text{Calculated}} \approx 2.6076$

• $Z_{\text{Tabulated/Critical}} = 1.96$

Since $2.6076 > 1.96$, the null hypothesis must be **rejected**.

- **Correct Result:** The calculated Z value is outside the acceptance region, so the Null Hypothesis ($H_0 : \mu = 3.2$) is **Rejected**.